
Chapter 2

Life Processes in Living Organisms – Part 1

1. Living Organisms and Life Processes

- In living organisms, many **organ systems** work continuously.
 - Important systems are:
 - Digestive system
 - Respiratory system
 - Circulatory system
 - Excretory system
 - Control system (nervous and hormonal)
 - All these systems work **independently but in coordination**.
 - Similar life processes occur in **all living organisms**.
 - All life processes need a **continuous supply of energy**.
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2. Energy Production in Living Organisms

- **Carbohydrates, fats and lipids** are the main sources of energy.
 - Energy is produced in the **mitochondria** present in each cell.
 - For energy production:
 - **Food** is necessary
 - **Oxygen** is also necessary
 - Food and oxygen are transported to cells by the **circulatory system**.
 - These processes are controlled by the **control system**.
 - Every life process contributes to **energy production**.
 - Energy is also required to perform all life processes.
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3. Autotrophs and Heterotrophs

- **Plants are autotrophs.**
 - They prepare their own food by photosynthesis.
- Plants use some food for themselves.
- Remaining food is stored in:

- Fruits
 - Leaves
 - Stem
 - Roots
 - Humans and animals consume:
 - Fruits
 - Vegetables
 - Plant products
 - From food, we obtain:
 - Carbohydrates
 - Proteins
 - Fats
 - Vitamins
 - Minerals
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4. Sources of Carbohydrates

We get carbohydrates from:

- Milk
- Fruits
- Jaggery
- Sugarcane sugar
- Vegetables
- Potatoes, sweet potatoes
- Sweet dishes
- Cereals like:
 - Wheat
 - Rice
 - Maize
 - Ragi
 - Jowar
 - Millet

👉 1 gram of carbohydrate gives 4 kcal of energy.

5. Why Do Players Eat During Breaks? (Use Your Brain Power)

- Players eat food during breaks to:
 - Get **instant energy**

- Reduce **tiredness**
 - Maintain **stamina and performance**
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6. Respiration in Living Organisms

- Respiration occurs at **two levels**:
 1. **Body level respiration**
 2. **Cellular level respiration**

Body Level Respiration

- Exchange of **oxygen and carbon dioxide** between body and surroundings.

Cellular Respiration

- Oxidation of food occurs in cells.
 - It may occur:
 - With oxygen → **Aerobic respiration**
 - Without oxygen → **Anaerobic respiration**
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7. Human Respiratory System (Diagram Question)

Label the diagram:

- Nasal cavity
- Trachea
- Bronchi
- Lungs
- Alveoli
- Diaphragm

(Important for exam diagram questions)

8. Glucose Molecule – Thinking Questions

1. Glucose contains:
 - Carbon (C) – 6 atoms

- Hydrogen (H) – 12 atoms
 - Oxygen (O) – 6 atoms
 - 2. Chemical bonds present:
 - Covalent bonds
 - 3. Oxidation in chemistry means:
 - Loss of electrons
 - Addition of oxygen
 - Release of energy
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9. Cellular Respiration

- Carbohydrates are mainly used for **energy production**.
- Energy is released in the form of **ATP (Adenosine Triphosphate)**.
- **Glucose** is oxidized step by step in cells.
- This process is called **cellular respiration**.

Types of Cellular Respiration:

1. Aerobic respiration (with oxygen)
 2. Anaerobic respiration (without oxygen)
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10. Aerobic Respiration – Steps

Step 1: Glycolysis

- Occurs in **cytoplasm**.
 - One glucose molecule produces:
 - 2 pyruvic acid molecules
 - 2 ATP
 - 2 NADH₂
 - Water
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Conversion of Pyruvic Acid

- Pyruvic acid is converted into **Acetyl Co-A**.
- During this process:
 - CO₂ is released

- NADH_2 is formed
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Step 2: Tricarboxylic Acid (TCA) Cycle / Krebs Cycle

- Occurs in **mitochondria**.
 - Acetyl Co-A enters the cycle.
 - Complete oxidation of glucose occurs.
 - Products formed:
 - CO_2
 - H_2O
 - NADH_2
 - FADH_2
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☆ Exam Tip:

- **Mitochondria = site of aerobic respiration**
 - **ATP = energy currency of cell**
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1. Mitochondria and Tricarboxylic Acid Cycle (TCA Cycle)

- **Mitochondria** is a double membrane organelle.
- It has:
 - Outer membrane
 - Inner membrane
 - Inner mitochondrial space (matrix)

👉 TCA cycle (Krebs cycle) occurs in the mitochondrial matrix.

👉 Pyruvic acid is converted into Acetyl-CoA and enters this cycle.

2. ATP – Energy Currency of the Cell

- ATP (Adenosine Triphosphate) is an energy-rich molecule.
- Energy is stored in the phosphate bonds of ATP.
- ATP is stored in cells as per need.

Structure of ATP:

ATP contains:

- Adenine (nitrogenous base)
- Ribose (pentose sugar)
- Three phosphate groups

ATP → ADP

- When energy is needed:
 - One phosphate bond breaks
 - ATP changes into ADP
 - Energy is released

👉 Therefore, ATP is called the “energy currency of the cell.”

3. Electron Transfer Chain Reaction (ETC)

- NADH₂ and FADH₂ are formed during:
 - Glycolysis
 - Conversion of pyruvic acid
 - Krebs cycle
- These molecules take part in electron transfer chain reaction.

ATP Formation:

- From 1 NADH₂ → 3 ATP molecules

- From 1 $\text{FADH}_2 \rightarrow 2$ ATP molecules
- Water molecules are also formed.
- ETC occurs **only in mitochondria**.

👉 In aerobic respiration:

- Glucose is completely oxidized
 - Final products:
 - CO_2
 - H_2O
 - Energy (ATP)
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4. NADH_2 and FADH_2

- NADH_2 – Nicotinamide Adenine Dinucleotide
 - FADH_2 – Flavin Adenine Dinucleotide
 - Both are **enzymes** formed in cells.
 - They help in **cellular respiration** and ATP production.
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5. Use of Fats and Proteins for Energy

- If carbohydrates are insufficient (fasting, hunger):
 - **Lipids** are converted into **fatty acids**
 - **Proteins** are converted into **amino acids**
 - Fatty acids and amino acids are converted into **Acetyl-CoA**.
 - Acetyl-CoA enters **Krebs cycle**.
 - Energy is released through complete oxidation.
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6. Scientists and Discoveries (Very Important for Exam)

Glycolysis:

4. Discovered by:
 - Gustav Embden
 - Otto Meyerhof
 - Jacob Parnas

👉 Hence, glycolysis is also called: **Embden–Meyerhof–Parnas (EMP) pathway**

Krebs Cycle:

- Discovered by **Sir Hans Krebs**
 - Also called **Krebs cycle**
 - He received the **Nobel Prize in 1953**
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7. Energy Production Pathway (Flow Chart Based)

For Carbohydrates:

Glucose → Glycolysis → Pyruvic acid → Acetyl-CoA → Krebs cycle → $\text{CO}_2 + \text{H}_2\text{O} + \text{Energy}$

For Fats:

Lipids → Fatty acids → Acetyl-CoA → Krebs cycle → Energy

For Proteins:

Proteins → Amino acids → Acetyl-CoA → Krebs cycle → Energy

8. Anaerobic Respiration in Microorganisms

3. Some organisms cannot live in presence of oxygen.
 1. Example: Many bacteria
4. These organisms perform **anaerobic respiration**.

Steps of Anaerobic Respiration:

- Glycolysis
 - Fermentation
 - Glucose is **incompletely oxidized**.
 - **Less energy** is produced.
-

9. Fermentation

- Pyruvic acid is converted into:
 - Alcohol (in yeast)
 - Organic acids (lactic acid, vinegar, etc.)
 - This process is called **fermentation**.
 - It occurs with the help of **enzymes**.
-

10. Anaerobic Respiration in Higher Organisms

- Some plants and animals also perform anaerobic respiration when oxygen is insufficient.

Examples:

1. Seeds perform anaerobic respiration when soil is waterlogged.
2. Muscle cells perform anaerobic respiration during heavy exercise.

Effect:

1. Less energy is produced
 2. **Lactic acid accumulates**
 3. This causes **muscle fatigue and tiredness**
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☆ Exam Key Points (Always Remember)

1. ATP = Energy currency of cell
 2. Mitochondria = Site of aerobic respiration
 3. Krebs cycle = Complete oxidation of glucose
 4. Anaerobic respiration = Less energy
 5. Lactic acid causes muscle fatigue
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1. Complete Oxidation of Glucose

Q1. Which type of cellular respiration performs complete oxidation of glucose?

👉 Aerobic respiration

Q2. Which cell organelle is necessary for complete oxidation of glucose?

👉 Mitochondria

2. Energy from Different Food Components

Storage of Carbohydrates

- Excess carbohydrates are stored in:
 - Liver
 - Muscles
 - Stored in the form of glycogen
-

3. Proteins

What are proteins made of?

- Proteins are **macromolecules**.
- They are made up of many **amino acids**.

Sources and Energy

- Proteins of animal origin are called **first class proteins**.
- **1 gram of protein gives 4 kcal of energy**.

Digestion and Use

- Proteins are digested into **amino acids**.
 - Amino acids are absorbed into blood.
 - They are transported to all organs and cells.
 - Cells use amino acids to form:
 - Enzymes
 - Hormones
 - Structural proteins
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4. Uses of Amino Acids in the Body

- **Skin** → Melanin, Keratin
 - **Bones** → Ossein
 - **Cells** → Cell membrane proteins, enzymes
 - **Pancreas** → Insulin, Trypsin
 - **Pituitary gland** → Hormones
 - **Muscles** → Actin and Myosin
 - **Blood** → Hemoglobin, Antibodies
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5. Excess Proteins and Amino Acids

- Excess amino acids are **not stored** in the body.
 - They are broken down.
 - **Ammonia** formed is removed from the body.
 - If needed, proteins are converted into **glucose** by **gluconeogenesis**.
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6. Proteins in Plants

- Plants produce amino acids from minerals **de novo**.
 - They synthesize different proteins.
 - **RUBISCO enzyme** present in chloroplasts is:
 - The **most abundant protein in nature**.
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7. Lipids (Fats)

What are lipids?

- Lipids are formed by chemical bonding of:
 - Fatty acids
 - Alcohol

Digestion of Lipids

- Lipids are digested into:
 - Fatty acids
 - Alcohol
- Fatty acids are absorbed and transported in the body.

Functions of Lipids

5. Formation of **phospholipids** for plasma membrane
6. Formation of hormones:
 - Estrogen

- Progesterone
 - Testosterone
 - Aldosterone
7. Formation of **myelin sheath** around nerve axons

Energy and Storage

- 1 gram of lipid gives 9 kcal of energy.
 - Excess lipids are stored in:
 - **Adipose connective tissue**
-

8. Vitamins

5. Vitamins are essential for proper functioning of the body.
6. Main types of vitamins:
1. A, B, C, D, E, K

Solubility:

- **Fat soluble** → A, D, E, K
 - **Water soluble** → B, C
 - Vitamins like:
 - Riboflavin (Vitamin B₂)
 - Nicotinamide (Vitamin B₃)
 - Are required for formation of:
 - FADH₂
 - NADH₂
-

9. Water – An Essential Nutrient

- Human body contains **65–70% water**.
- Each cell contains **70% water**.
- Blood plasma contains **90% water**.

- Even small loss of water disturbs body functions.
 - Therefore, **water is an essential nutrient.**
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10. Dietary Fibers

- Fibers cannot be digested.
 - They help in:
 - Digestion
 - Egestion of undigested food
 - Sources:
 - Leafy vegetables
 - Fruits
 - Cereals
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11. Cell Division – An Essential Life Process

- Cell division helps in:
 - Healing of wounds
 - Growth of organisms
 - Formation of new individuals
 - Repair of damaged tissues

Types of Cell Division:

6. Mitosis

1. Occurs in somatic and stem cells
2. Helps in growth and repair

7. Meiosis

1. Occurs in germ cells
 2. Helps in reproduction
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☆ Exam Key Points

8. Aerobic respiration → complete oxidation
9. Mitochondria → energy production
10. Glycogen → stored carbohydrate
11. Proteins → body building nutrients
12. Lipids → highest energy source
13. Vitamins → regulate body processes
14. Water & fibers → essential nutrients
15. Cell division → growth, repair, reproduction

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1. Chromosome (2.5)

- Chromosomes carry genetic material (DNA).
- In **diploid (2n)** condition:
 - A pair of each type of chromosome is present.
- In **haploid (n)** condition:
 - Only one chromosome of each type is present.
- Structure of chromosome:
 - Two sister chromatids
 - Joined at centromere

2. Mitosis (2.6)

- Mitosis occurs in:

- Somatic cells
 - Stem cells
- It helps in:
 - Growth of body
 - Wound healing
 - Repair of tissues
 - Formation of blood cells

Steps of Mitosis:

Mitosis occurs in **two main stages**:

- **Karyokinesis** (nuclear division)
 - **Cytokinesis** (cytoplasmic division)
-

3. Karyokinesis – Phases

A. Prophase

- Chromosomes condense and become **short and thick**.
 - Sister chromatids become visible.
 - Centrioles duplicate and move to **opposite poles**.
 - Nuclear membrane and nucleolus start disappearing.
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B. Metaphase

- Nuclear membrane **completely disappears**.
 - Chromosomes become clearly visible.
 - Chromosomes align at the **equatorial plane**.
 - **Spindle fibers** connect centromere and centrioles.
-

C. Anaphase

- Centromeres split.
 - Sister chromatids separate.
 - They move to **opposite poles**.
 - Separated chromatids are called **daughter chromosomes**.
 - Chromosomes appear like a **bunch of bananas**.
-

D. Telophase

- Chromosomes reach opposite poles.
 - They decondense and become thin.
 - Nuclear membrane reforms.
 - Nucleolus reappears.
 - Two daughter nuclei are formed.
 - Spindle fibers disappear.
-

4. Cytokinesis (2.7)

- Cytoplasm divides to form **two daughter cells**.

In animal cells:

- A **notch** forms at the equator and deepens.

In plant cells:

8. A **cell plate** forms at the center.
-

5. Importance of Mitosis

- Growth of multicellular organisms
- Repair of damaged tissues

- Wound healing
 - Replacement of worn-out cells
-

6. Meiosis (2.8 & 2.9)

7. Meiosis occurs in **germ cells**.
 8. It is completed in **two stages**:
 1. **Meiosis-I**
 2. **Meiosis-II**
-

Meiosis-I

- Homologous chromosomes pair up.
 - **Crossing over** (exchange of genetic material) occurs.
 - Homologous chromosomes separate.
 - Two **haploid (n)** cells are formed.
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Meiosis-II

- Similar to mitosis.
 - Sister chromatids separate.
 - Four **haploid daughter cells** are formed.
-

7. Importance of Meiosis

- Formation of **gametes and spores**
 - Maintains chromosome number in species
 - Causes **genetic variation**
 - Daughter cells are genetically different
-

8. Definitions – Very Important for Exam

Diploid cell (2n):

- Cell having **two sets of chromosomes**.

Haploid cell (n):

16. Cell having **one set of chromosomes**.

Homologous chromosomes:

17. Chromosome pairs with:
1. Same shape
 2. Same size
 3. Same gene position

Gametes:

18. **Haploid**, because they contain single set of chromosomes.
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9. Onion Root Tip Experiment (Practical)

1. Used to observe **mitosis**.
2. Rapidly dividing cells are present in root tips.

Observation:

1. Different stages of mitosis can be seen:
 1. Prophase
 2. Metaphase
 3. Anaphase
 4. Telophase
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☆ Exam Key Points

1. Mitosis → growth and repair
2. Meiosis → gamete formation
3. Crossing over → genetic variation
4. 1 diploid cell → 4 haploid cells (meiosis)
5. Onion root tip → mitosis study